Unit 4.01: Conditional Probability & Bayes' Rule

Q1. In a dataset, the prior probability of class C_1 is PC_1 = 0.10 . What is the probability of the complement class Pneg C_1 ?

[x] A: 0.90 [ ] B: 0.95
[ ] C: 0.05 [ ] D: 1.10

Solution: By the complement rule of probability: Pneg C_1 = 1 - PC_1 = 1 - 0.10 = 0.90 .

Q2. In a dataset, the prior probability of class C_1 is PC_1 = 0.20 . What is the probability of the complement class Pneg C_1 ?

[x] A: 0.80 [ ] B: 0.90
[ ] C: 0.10 [ ] D: 1.20

Solution: By the complement rule of probability: Pneg C_1 = 1 - PC_1 = 1 - 0.20 = 0.80 .

Q3. In a dataset, the prior probability of class C_1 is PC_1 = 0.60 . What is the probability of the complement class Pneg C_1 ?

[x] A: 0.40 [ ] B: 0.70
[ ] C: 0.30 [ ] D: 1.60

Solution: By the complement rule of probability: Pneg C_1 = 1 - PC_1 = 1 - 0.60 = 0.40 .

Q4. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.05 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[x] A: 0.0450 [ ] B: 0.9500
[ ] C: 0.8500 [ ] D: 0.0050

Solution: By the multiplication rule of conditional probability: Ptext{Defect} cap text{Positive} = Ptext{Defect} times Ptext{Positive} | text{Defect} = 0.05 times 0.90 = 0.0450 .

Q5. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.10 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[x] A: 0.0900 [ ] B: 1.0000
[ ] C: 0.8000 [ ] D: 0.0100

Solution: By the multiplication rule of conditional probability: Ptext{Defect} cap text{Positive} = Ptext{Defect} times Ptext{Positive} | text{Defect} = 0.10 times 0.90 = 0.0900 .

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Q6. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.15 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[x] A: 0.1350
[ ] B: 1.0500
[ ] C: 0.7500
[ ] D: 0.0150

Solution: By the multiplication rule of conditional probability: Ptext{Defect} cap text{Positive} = Ptext{Defect} times Ptext{Positive} | text{Defect} = 0.15 times 0.90 = 0.1350 .

Q7. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.20 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[x] A: 0.1800
[ ] B: 1.1000
[ ] C: 0.7000
[ ] D: 0.0200

Solution: By the multiplication rule of conditional probability: Ptext{Defect} cap text{Positive} = Ptext{Defect} times Ptext{Positive} | text{Defect} = 0.20 times 0.90 = 0.1800 .

Q8. A machine learning diagnostic test has sensitivity Ptext{Positive} | text{Defect} = 0.90 . If the defect rate in production is Ptext{Defect} = 0.25 , find the joint probability of a defect and a positive test result Ptext{Defect} cap text{Positive} .

[x] A: 0.2250
[ ] B: 1.1500
[ ] C: 0.6500
[ ] D: 0.0250

Solution: By the multiplication rule of conditional probability: Ptext{Defect} cap text{Positive} = Ptext{Defect} times Ptext{Positive} | text{Defect} = 0.25 times 0.90 = 0.2250 .